

Evan Smith

Final Project Proposal

What is the value of a NHL draft pick?

For my final project, I would like to create a program to assign and predict values for draft picks for the National Hockey League. This program should take the statistics of different players, their position and their draft spot/round. I would like to show how valuable an early fourth round pick is in comparison to an early third round pick. Additionally, I would like to see if there is a golden spot for earlier goalie selections, a question that has plagued teams for years.

I intended to get a large sum of the data, in particular draft data, from <https://www.eliteprospects.com/draft/nhl-entry-draft/2021>. Additionally, I am mainly writing to check if I can use data downloaded from <https://moneypuck.com/data.htm>, which allows the downloads of Excel files for skaters, goalies, line pairs, etc. This excel data is useful as it contains very specific pieces of recorded data I cannot find elsewhere, such as high danger shot counts.

The Excel files will require duplication checks as many of the files repeat players to show the larger variety of stats, and I will need to map players to their correct stats from the eliteprospects which will require additional checks as many active players have the same name. Analysis wise I am going to check different sklearn libraries to find the best way to weigh different stats and generate appropriate scores based on stats for draft pick numbers and rounds. I will try to determine the number of “successful” picks per position, per round based on stats such as points, games played, goals allowed, etc., though each position will require their own category choices and weights.

I will attempt to visualize this with scatter plot diagrams to show progressive changes or sharp turns along with bar plots to show the number of “successful” picks per round. I hope this proposal is ok and look forward to working on it!